

Experiment resistivity of a wire

Aim To find the resistivity of the material of a given wire using a metre bridge

Apparatus metre bridge galvanometer resistance box jockey battery one way key and thick connecting wires

Theory The resistance of a uniform wire is proportional to its length and inversely proportional to its area of cross section the constant of proportionality is the resistivity of the material

Result the resistivity was found to be constant for all the lengths tested within the limits of experimental error